

Inpatient Evaluation of Acute Dyspnea

Objectives

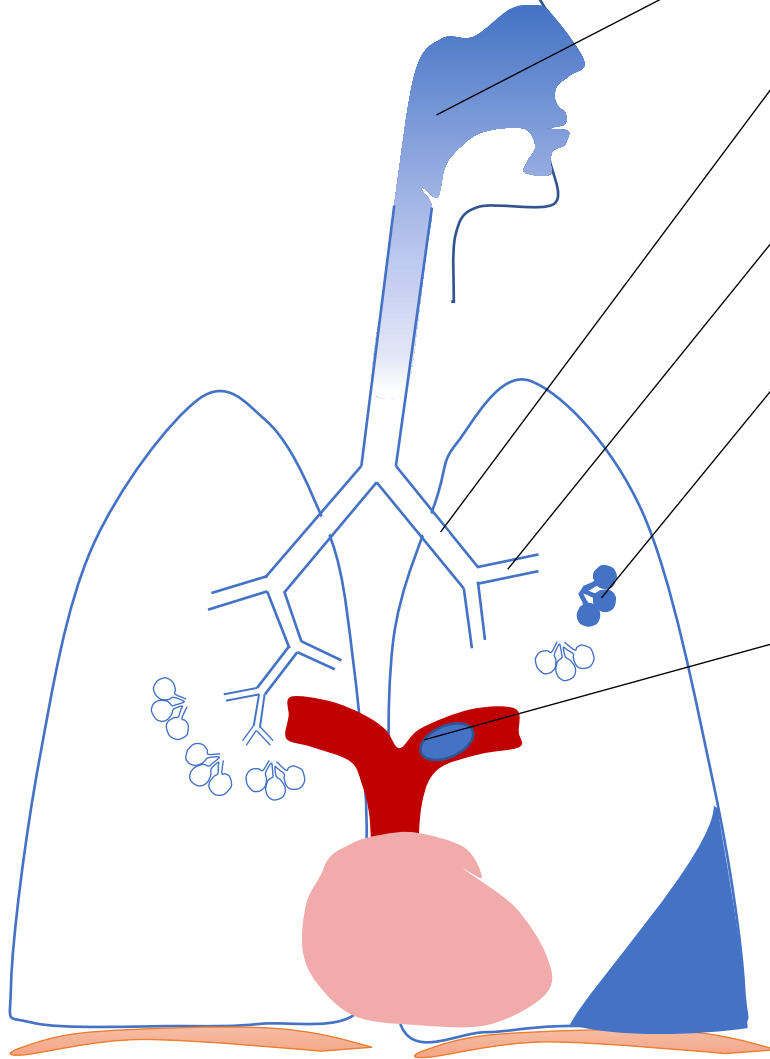
1. Develop a broad differential for causes of acute dyspnea

2. Construct a framework for the evaluation of acute dyspnea

3. Identify causes of acute hypoxemia and how to differentiate them

PRACTICE CASES

1. Causes of Acute Dyspnea



Airways

Anaphylaxis, angioedema
Mucus plugging, airway stenosis
Bronchoconstriction (COPD, asthma)

Alveoli

<u>Fills alveoli</u>	<u>Collapses alveoli</u>
Blood = DAH	Atelectasis
Pus = pneumonia	Pleural process (blood, pus, water, air)
Water = edema	

Vessels

Pulmonary embolism

Diaphragm dysfunction
Abdominal pressure

Myocardial infarction
Arrhythmias Pericardial disease
Valvular disease

Anxiety
Acidosis (metabolic and respiratory)
Anemia

RESPIRATORY

Click here

Blockage or narrowing of any part of the respiratory system will cause dyspnea!

MUSCULOSKELETAL

CARDIAC

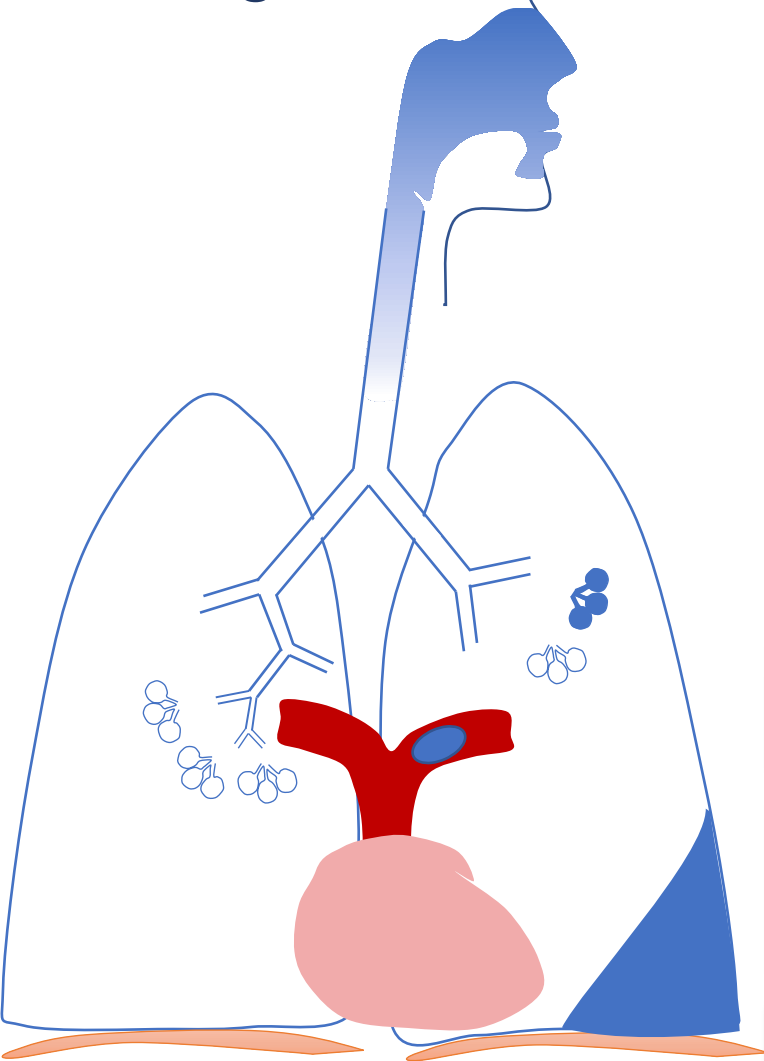
The 3 "A's"



2 EVAL

3 HYPOXEMIA

2. Evaluation and Diagnostics



1 CAUSES

3 HYPOXEMIA

Airways
Anaphylaxis, angioedema
Mucus plugging, airway stenosis
Bronchoconstriction (COPD, asthma)

Alveoli

Fills alveoli
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Pus = pneumonia
Water = edema

Collapses alveoli
Atelectasis
Pleural process (blood, pus, water, air)

Vessels
Pulmonary embolism

Diaphragm dysfunction
Abdominal pressure

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Arrhythmias
Valvular disease

Anxiety
Acidosis (metabolic and respiratory)
Anemia

How to Evaluate

Physical exam

CXR

CTPE

CXR, ABG/ VBG
Physical exam

ECG

Physical exam

BMP, ABG/ VBG
CBC

ABG

More Eval

2. Evaluation and Diagnostics

Important diagnostic tests:

CXR

ECG

CBC

ABG vs. VBG

The Age-Old Question: ABG or VBG?

ABG: pH / $P_a\text{CO}_2$ / $P_a\text{O}_2$

VBG: pH / $P_v\text{CO}_2$

ABG pH = VBG pH + 0.04
ABG $P_a\text{CO}_2$ = VBG $P_v\text{CO}_2$ - 4

Get an ABG when you need the $P_a\text{O}_2$:

1. Calculate the A-a gradient
2. Confirming hypoxemia
3. Assessing the severity of hypoxemia

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1 CAUSES

3 HYPOXEMIA

Airways

Anaphylaxis, angioedema
Mucus plugging, airway stenosis
Bronchoconstriction (COPD, asthma)

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Fills alveoli

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How to Evaluate

Physical exam

CXR

CTPE

CXR, ABG/ VBG
Physical exam

ECG

Physical exam

BMP, ABG/ VBG
CBC

ABG

3. Hypoxemia

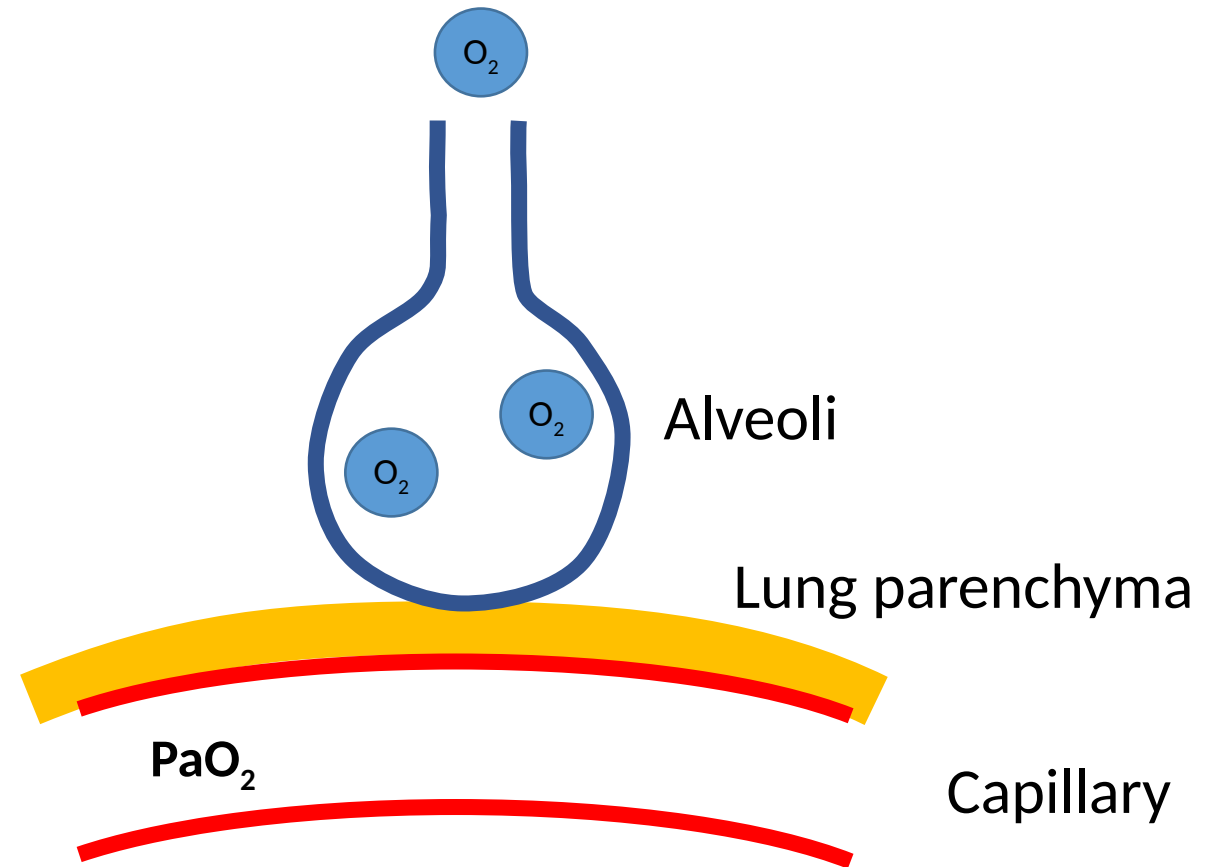
Dyspnea $\neq$ Hypoxemia

PaO_2 is measured on an ABG.

Normal PaO_2 is 75 -100.

Hypoxemia = low PaO_2

Not all causes of dyspnea cause hypoxemia.



Click Anywhere to Advance
Slide Animations and Text

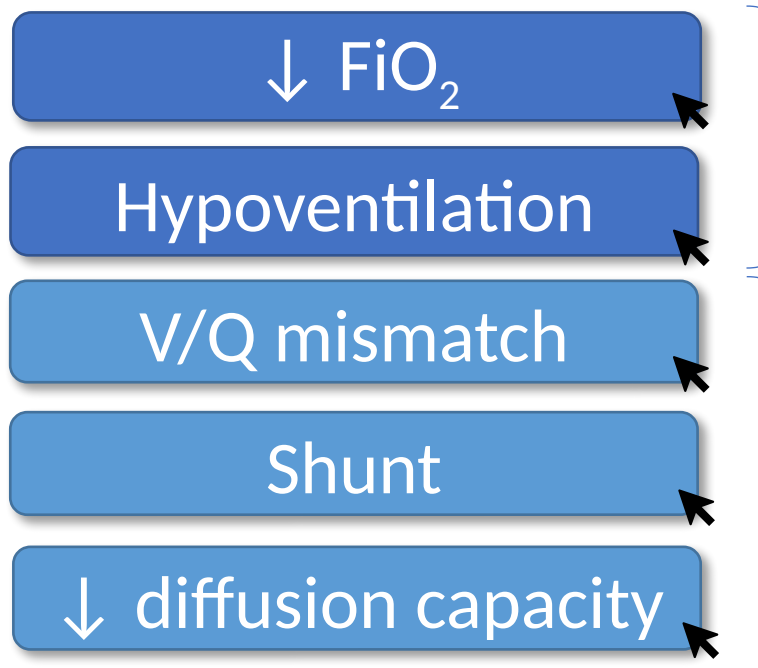


1 CAUSES

2 EVAL

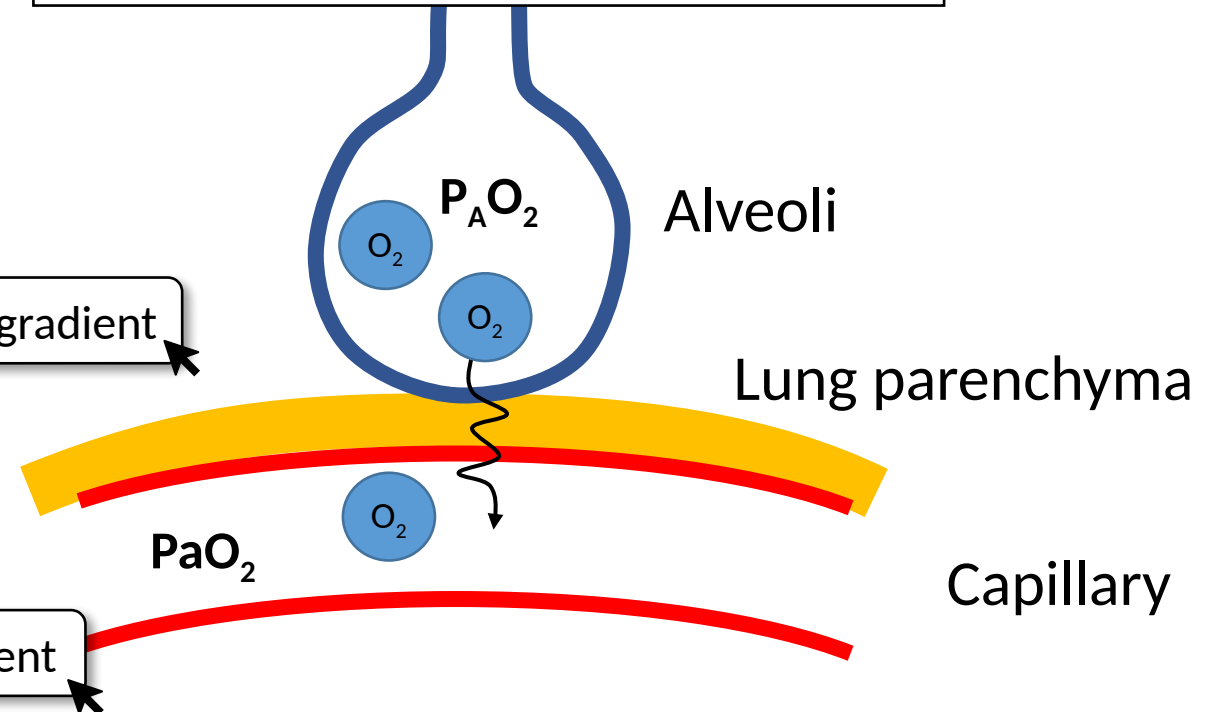
Causes of Hypoxemia

There are 5 causes of hypoxemia.



Normal A-a gradient

↑ A-a gradient



An elevation in the ventilatory rate	The A-a gradient = $P_A O_2 - PaO_2$ $P_A O_2 = FiO_2 \times (P_{atm} - P_{H_2O}) - P_a CO_2 (1.25)$ PaO_2 and PCO_2 are obtained from an ABG The expected A-a gradient = $age / 4 + 4$	option in (aka ion) can t.
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1 CAUSES

2 EVAL

Hypoventilation

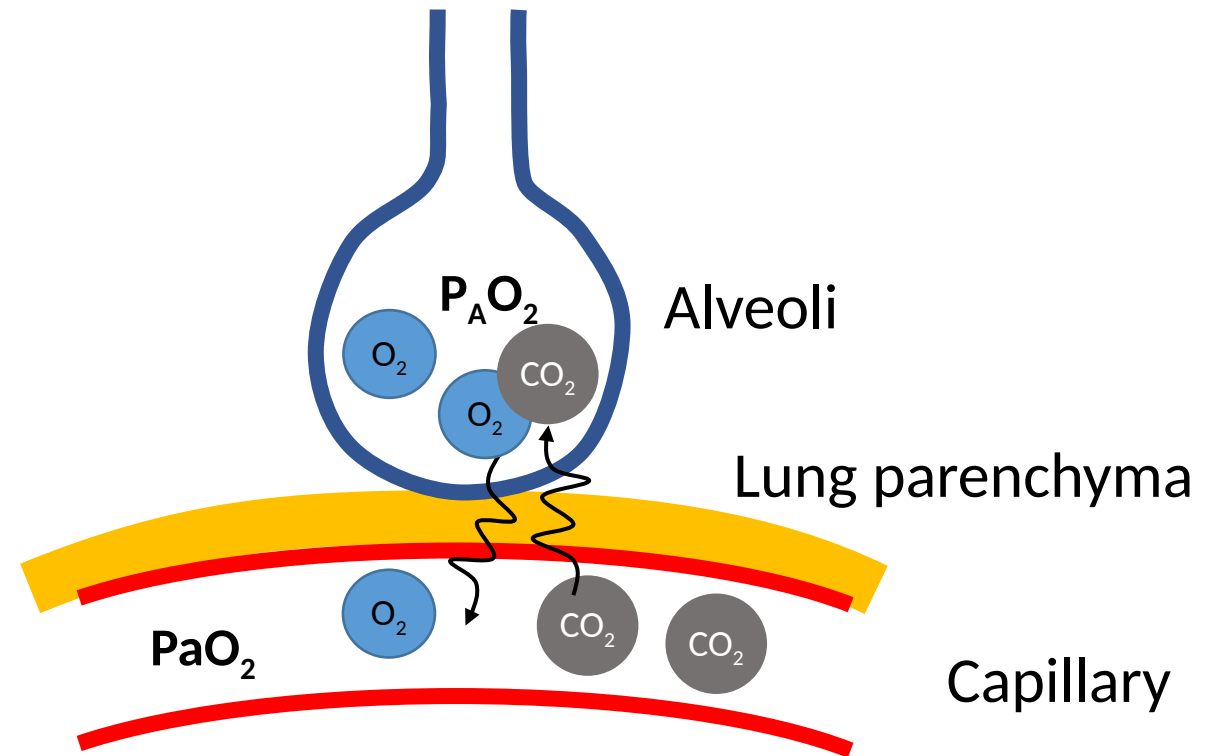
↓ FiO_2

Hypoventilation

V/Q mismatch

Shunt

↓ diffusion capacity



Ventilation, which is determined by respiratory rate and tidal volume (volume of breaths) creates opportunities for gas exchange. Hypoventilation ↓ elimination of CO_2 resulting in ↑ PCO_2 . Elevated PCO_2 effectively ↓ “space” for O_2 , and subsequently results in hypoxemia. Examples include CNS depression, opiates/ benzodiazepines, neuromuscular weakness, diaphragmatic dysfunction.



1 CAUSES

2 EVAL

V/Q mismatch

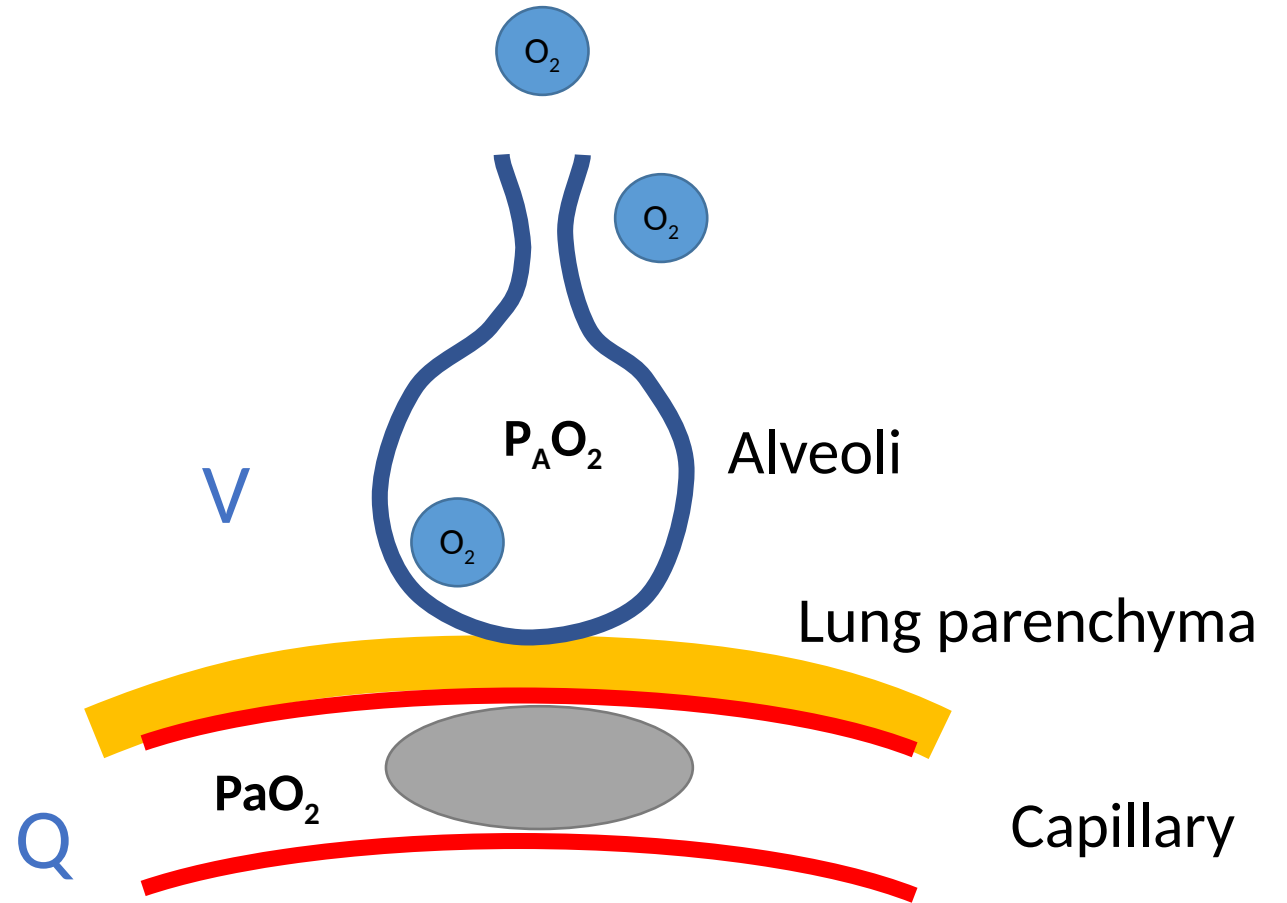
↓ FiO_2

Hypoventilation

V/Q mismatch

Shunt

↓ diffusion capacity



Ventilation (V) - Perfusion (Q) mismatch (or V/Q mismatch) occurs when there are areas of V without Q or vice versa. This can occur in the presence of smaller alveolar filling/collapsing process, obstructive airway disease, pulmonary vascular disease, or pulmonary embolism.



1 CAUSES

2 EVAL

Shunt

↓ FiO_2

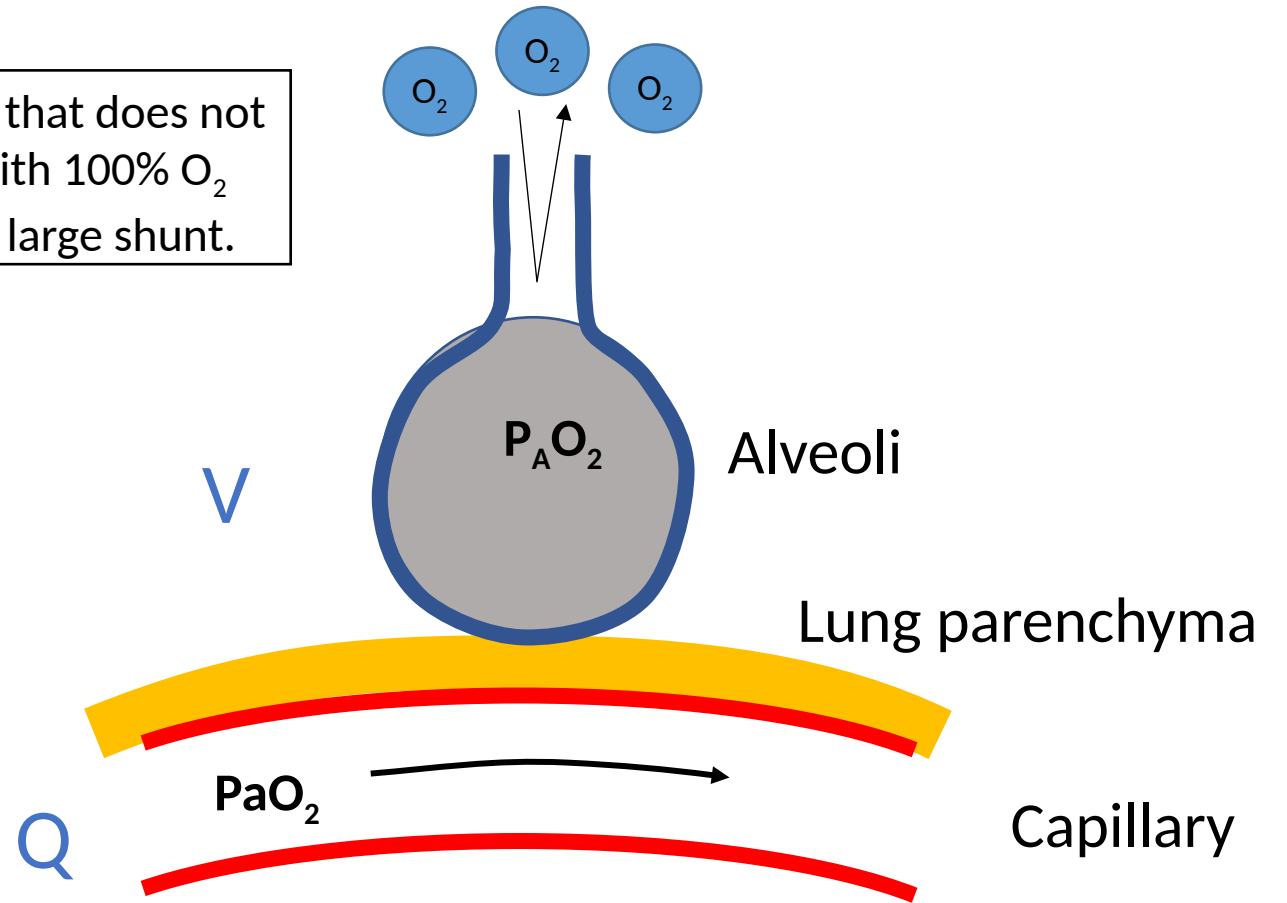
Hypoventilation

V/Q mismatch

Shunt

↓ diffusion capacity

Hypoxemia that does not correct with 100% O_2 suggests a large shunt.



Shunt occurs when there is perfusion (Q) without ventilation (V). This can occur in the presence of a physical shunt or with a large alveolar filling/collapsing process. Things that fill the alveoli include blood, water, pus.



1 CAUSES

2 EVAL

CASES

CASE 1

84 M with a history of HLD, CAD presents with 1 d of dyspnea.

- VS: T 38C, HR 110, BP 125/70, RR 24, S_aO₂ 92% on RA

• Ex

CXR looks like...



CXR

BMP/ CBC

ABG/ VBG

ECG

FINAL DIAGNOSIS

CASE 2

74 F with a history of NHL presents with 1 d of dyspnea.

- VS: T 36C, HR 110, BP 98/60, RR 24, S_aO₂ 94% on RA

• Ex

CXR looks like...



CXR

BMP/ CBC

ABG/ VBG

ECG

FINAL DIAGNOSIS

CASE 3

25 F with type 1 DM a presents with 1 d of dyspnea.

- VS: T 36C, HR 110, BP 125/70, RR 32, S_aO₂ 99% on RA

• Exam:

- Tachycardic, regular rhythm
- Tachypneic without increased work of breathing, clear lungs

Diagnostic testing

CXR

BMP/ CBC

ABG/ VBG

ECG

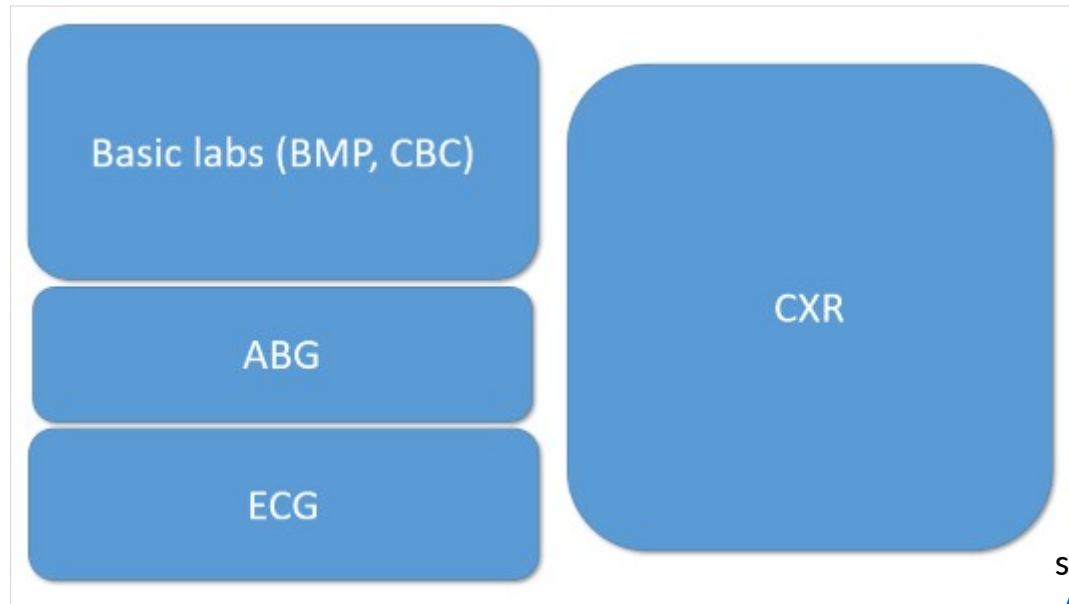
FINAL DIAGNOSIS

CASE 4 (part 1)

64 M with a history of R-sided renal cell cancer admitted for a R nephrectomy.

On post-op day 1, he developed acute onset shortness of breath and chest and abdominal pain.

- VS: T 37.1C, HR 140, BP 90/60, RR 28, S_aO_2 96% on RA
- Tachypneic, diminished lung sounds at bases. Abdomen distended and tender



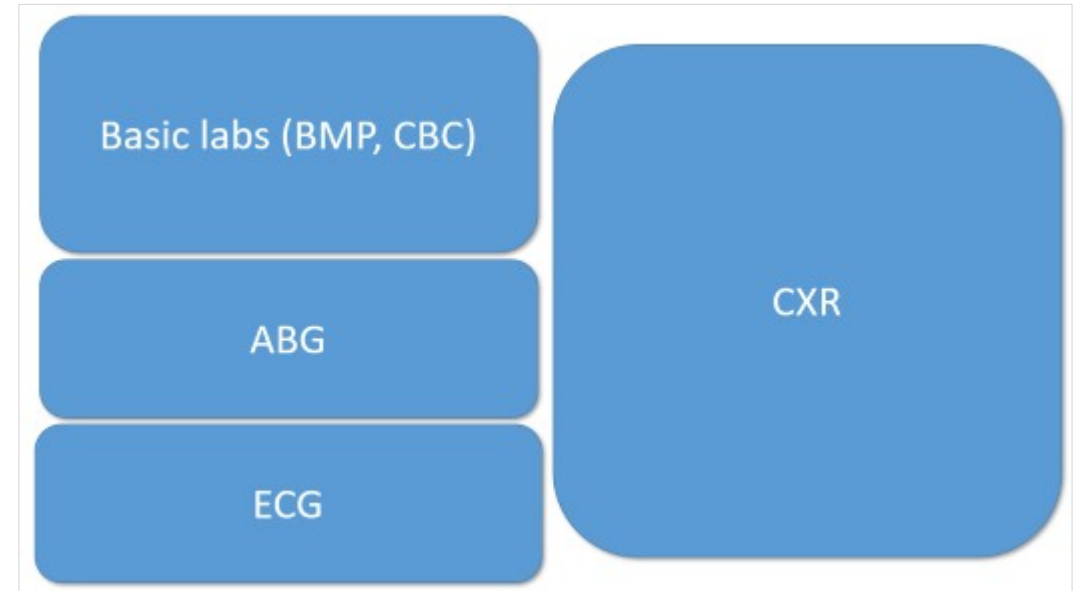
FINAL DIAGNOSIS

shows evi
[A-a gradi](#)

CASE 4 (part 2)

On post-op day 5, he developed acute onset shortness of breath and chest pain.

- VS: T 37.1C, HR 140, BP 90/60, RR 28, S_aO_2 82% on RA
- Tachypneic, lungs otherwise clear. Tachycardic without murmurs.



FINAL DIAGNOSIS



Basic labs (BMP, CBC)

ABG

PO₂, Z₁

ECG

CXR

Basic labs (BMP, CBC)

ABG

ECG

CXR

Calculate the A-a gradient

pH 7.43 / PCO₂ 32 / PO₂ 50 / HCO₃ 24 (on room air)

Calculate an A-a gradient!

$$P_AO_2 = 150 - 32 \times (1.25)$$

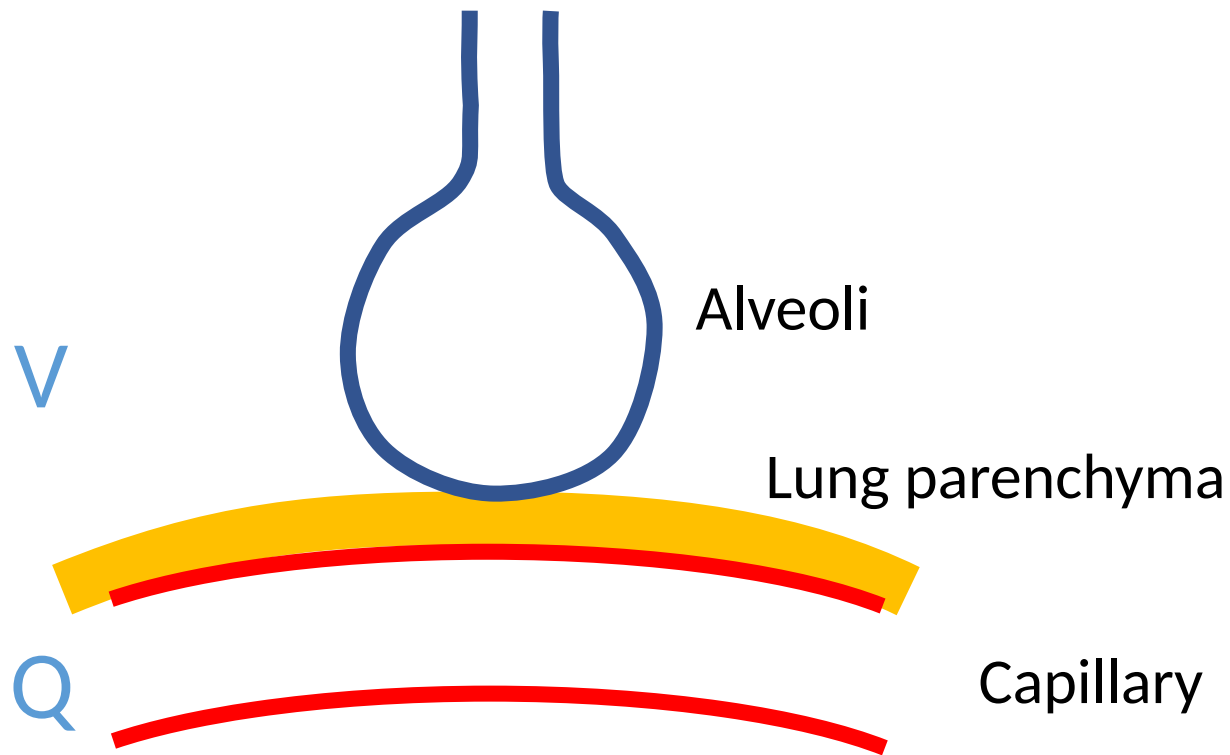
$$P_AO_2 = 110$$

$$P_AO_2 - P_aO_2 = 110 - 50 = 60$$

Elevated A-a gradient!

$$\begin{aligned}\text{Expected A-a gradient} &= \text{Age}/4 + 4 \\ &= 64/4 + 4 \\ &= 20\end{aligned}$$

Causes of an elevated A-a gradient



~~Shunt (anatomic, large alveolar filling process)~~

~~V/Q mismatch (obstructive lung disease, pulmonary vascular disease/PE)~~

~~↓ diffusion capacity~~

Non-anatomic shunts
are evident on CXR

Does not typically
happen acutely

CXR looks
like...



Image from Radiopaedia.org

GO BACK

The Age Old Question: ABG or VBG?

ABG: pH / $P_a\text{CO}_2$ / $P_a\text{O}_2$

VBG: pH / $P_v\text{CO}_2$

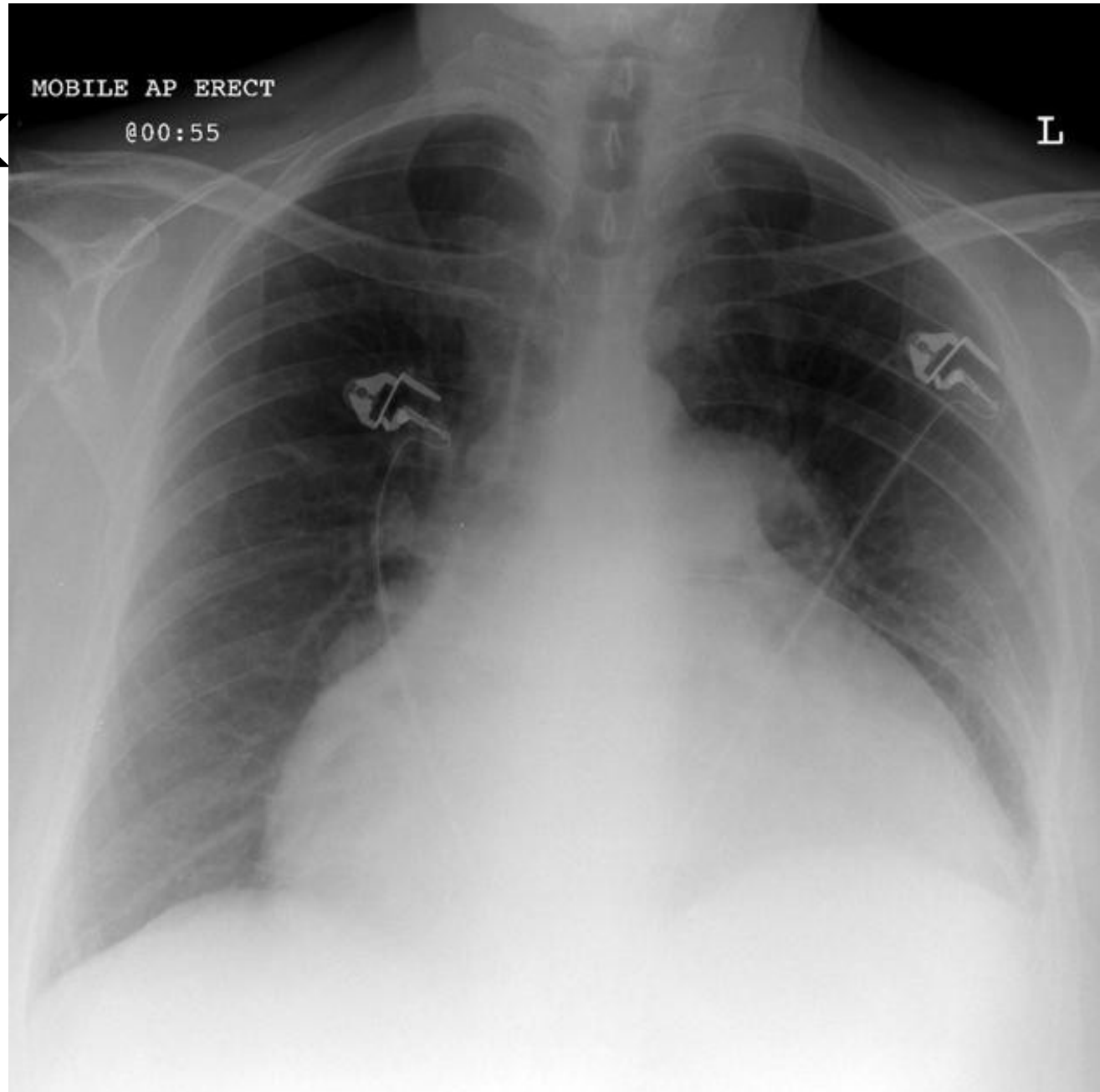


ABG pH = VBG pH + 0.04
ABG $P_a\text{CO}_2$ = VBG $P_v\text{CO}_2$ - 4

Get an ABG when you need the $P_a\text{O}_2$:

1. Calculate the A-a gradient
2. Confirming hypoxemia
3. Assessing the severity of hypoxemia

CXR looks like



GO BACK

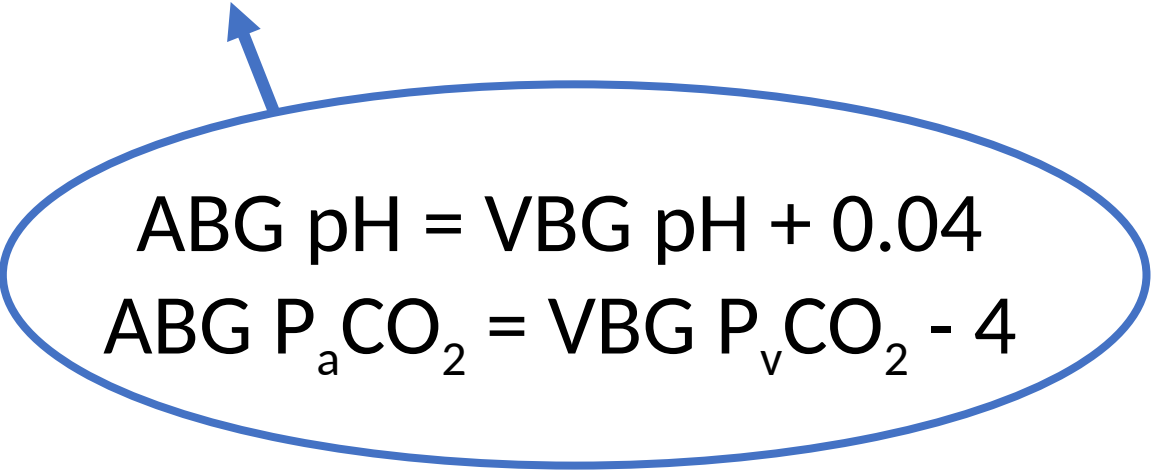
The Age-Old Question: ABG or VBG?

ABG: pH / $P_a\text{CO}_2$ / $P_a\text{O}_2$

VBG: pH / $P_v\text{CO}_2$

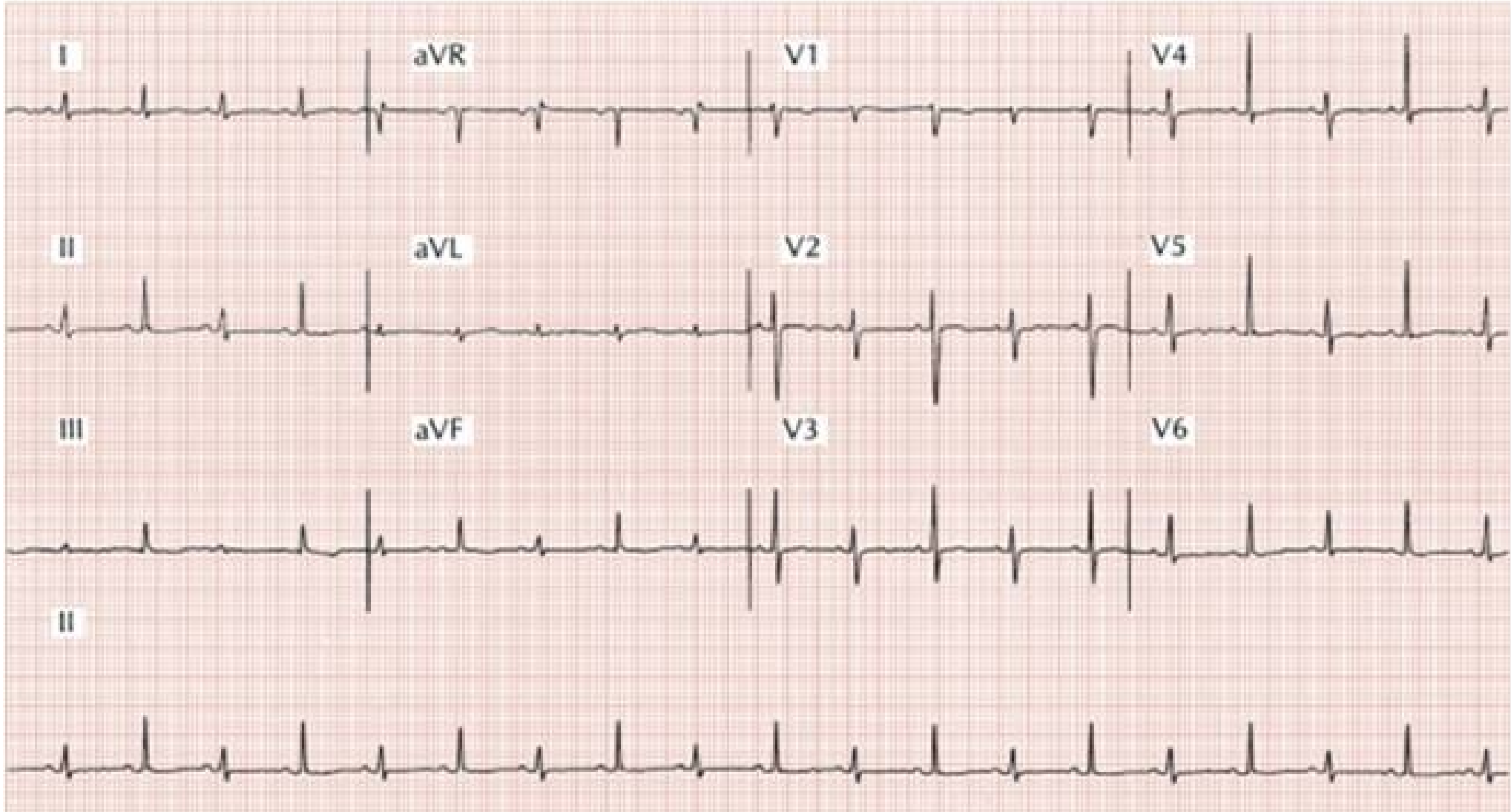
Get an ABG when you need the $P_a\text{O}_2$:

1. Calculate the A-a gradient
2. Confirming hypoxemia
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$$\begin{aligned}\text{ABG pH} &= \text{VBG pH} + 0.04 \\ \text{ABG } P_a\text{CO}_2 &= \text{VBG } P_v\text{CO}_2 - 4\end{aligned}$$

ECG looks like...

[GO BACK](#)



Acute Dyspnea / Hypoxemia — Framework Notes

The differential, the evaluation toolbox, and the 5-cause / A-a-gradient framework are timeless physiology, verified current (June 2026). Notes below reinforce, not replace, the standing teaching.

Two equations to keep on the board: $A-a \text{ gradient} = PAO_2 - PaO_2$; $PAO_2 = FiO_2 \times (P_{atm} - P_{H_2O}) - PaCO_2/0.8$ ($\approx 150 - 1.25 \times PaCO_2$ on room air). Expected A-a = $age/4 + 4$ — so a 64-year-old expects ~ 20 , not 16.

5 causes of hypoxemia: low FiO_2 and low diffusion capacity (NORMAL A-a, and not acute inpatient causes), then hypoventilation (normal A-a, high $PaCO_2$), V/Q mismatch, and shunt (both ELEVATED A-a). Shunt is the one that does NOT correct with 100% O_2 .

Point-of-care lung ultrasound (BLUE protocol; Lichtenstein) is now a routine bedside adjunct — rapidly distinguishes cardiogenic edema, pneumonia/consolidation, pneumothorax, and effusion with high sensitivity, complementing CXR/ABG/ECG/CBC.

ABG vs VBG: get an ABG when you need the PaO_2 (to confirm/grade hypoxemia or compute the A-a gradient). VBG estimates: $ABG \text{ pH} \approx VBG \text{ pH} + 0.04$; $ABG \text{ PaCO}_2 \approx VBG \text{ PvCO}_2 - 4$.

Take-Home Points

1. Build the dyspnea differential by system: respiratory (airways / alveoli / vessels), cardiac, musculoskeletal, and the "3 A's" (anemia, anxiety, acidosis). Respiratory and cardiac are most common.
2. The core work-up toolbox: CXR, blood gas, CBC, ECG — add POCUS at the bedside.
3. Dyspnea $\neq$ hypoxemia. Hypoxemia = low PaO₂ (normal 75–100); confirm it on an ABG.
4. Sort hypoxemia by the A-a gradient: normal (hypoventilation, low FiO₂) vs elevated (V/Q mismatch, shunt; low diffusion is chronic). Shunt fails to correct with 100% O₂.
5. Get an ABG over a VBG whenever you need the PaO₂ — e.g., to calculate the A-a gradient.

Attribution & Optimization Notes

Original lesson: "Inpatient Evaluation of Hypoxemia," TeachIM.org (updated Dec 2022). Two interactive boards published: Teaching (Presenter) and Learning (Learner).

Author: Yilin Zhang MD (expert review pending, per the lesson page).

Source URL: teachim.org/teaching_material/inpatient-evaluation-of-hypoxia/

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Correction made on BOTH boards (single-run XML, animations preserved): Case 4 expected A-a gradient " $= \text{Age}/4 + 4 = 64/4 + 4 = 16$ " -> " $= 20$ " ($64/4 + 4 = 20$; the measured 60 is still elevated, conclusion unchanged).

Human review: Case 3 (tamponade) CXR line reads "boot shaped heart which could indicate cardiomegaly or PLEURAL effusion" — given elevated JVP, electrical alternans, and the pericardial-effusion/tamponade diagnosis, this likely should read "pericardial effusion." Left unchanged pending author confirmation (also: "boot-shaped" is itself atypical wording for pericardial effusion).

Companions copied & linked: Learner board (SOB-Learner-Final.pptx, same A-a fix) + handout (SOB-worksheet.pdf, reviewed unchanged). Context source-cited: BLUE-protocol lung US (Lichtenstein). Medical content should be clinician-reviewed before teaching.